Code-Guided Video Editing with Source-Grounded Executable Proxies

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Figure 1. **An executable plan for editing an existing video.** Given the source and an instruction to add a wall clock, a coding agent builds a procedural scene that makes the intended placement explicit. A frozen video model combines this proxy with the source video to produce the edit. The no-proxy result adds an oversized clock; proxy guidance follows the planned placement and scale more closely. All four views are taken at the same time in the wall clock addition example.

Abstract

Video editing requires translating a user's intent into coordinated changes to objects, scene layout, camera framing, and temporal behavior while retaining the relevant content of the source video. We formulate this task as a transition between executable scene programs: Video2Code *estimates a source scene,* next-code prediction *expresses the requested intervention as its successor program, and* Code2Video *realizes the edited program through video synthesis. This separation exposes both a source-grounded scene state and an editing operation, with an executable preview that can be inspected before synthesis. Our current training-free prototype jointly realizes the first two logical stages with a video-aware coding agent and renders the edited program into a proxy for a frozen reference-conditioned video generator. The source video remains available throughout sampling, while the edited proxy participates only in the early sampling steps. Selected examples from all eight OpenVE-Bench editing categories illustrate object addition and removal, local changes, background replacement, camera reframing, stylization, creative transformations, and subtitle removal. Matched no-proxy comparisons highlight improvements in edit localization, source-content retention, and target framing in these examples. These examples establish a practical interface between programmatic editing and generative video synthesis. The program-transition view also defines a route to structured reconstruction and editing data for subsequent model training.*

1. Introduction

Editing a video often begins with a short instruction: add a bird in the upper-left sky, remove a person from a room, or widen the view of a clock tower. Each request implies a concrete intervention in a scene. An added object needs a location and behavior over time; a removal needs a plausible account of the newly visible background; a framing change needs a camera plan. At the same time, the source footage supplies content that the creator wants to retain, including the identity of the scene, the subject's performance, and the timing of the shot. An effective editing interface should make the intended intervention explicit while continuing to use this source information.

Diffusion-based visual synthesis has developed powerful generative priors [10, 22]. Video editing methods build on these priors through attention reuse, feature propagation, reference conditioning, and learned editing models [7, 12, 14, 21]. These advances make a broad range of edits possible. They also motivate a complementary question: what representation should a creator or an agent use to specify the target change before the final video is synthesized? For edits involving layout, object relations, or cam-

era framing, an executable representation offers a direct way to express and inspect the plan.

A scene program can instantiate objects, assign their attributes, define a camera, and evaluate their states at a chosen time. Rendering that program produces a visual preview of the requested edit. The preview makes the relation between an instruction and its spatial and temporal consequences visible. For example, the added clock in Fig. 1 has an explicit position relative to the person and the wall before the final video is generated. The program also remains available for inspection and revision, providing an artifact alongside the rendered result.

We organize this interface as *Video2Code* $\rightarrow$ *next-code prediction* $\rightarrow$ *Code2Video*. The first operation estimates a procedural account of the observed source. The second predicts the next program state under the editing instruction: objects can be inserted or removed, attributes changed, and cameras reframed within the same executable representation. Here, "next" denotes the edited scene program, rather than the next video frame. The third operation renders this program and uses its proxy to guide a reference-conditioned video generator. The resulting division assigns source reconstruction, explicit intervention, and visual realization separate roles. The original footage remains available for appearance and performance that the procedural scene only approximates.

The current training-free prototype gives the source and instruction jointly to a coding agent, which directly emits the edited program. It therefore realizes reconstruction and program editing jointly, without storing a separate pre-edit program. The edited program is executed, inspected, and refined before synthesis. We preserve the source-video condition throughout generation and provide the rendered proxy only during the initial sampling steps.

Our implementation builds on Code Video Model's executable video previews and early-step conditioning [15]. We adapt this foundation to source-video editing by grounding program construction in observed footage and retaining that footage as a persistent reference during synthesis. The current system requires no additional training of the video generator. We present successful examples drawn from the eight editing categories of OpenVE-Bench [8]. The examples expose the source–edited-program–proxy–output chain and illustrate how one executable interface can express varied editing requests. This technical report focuses on the method and its qualitative applications; it also outlines a path from the current system to structured editing data and future learned models.

Our contributions are:

- A three-stage formulation of source-video editing as scene-program reconstruction, instruction-conditioned next-code prediction, and source-grounded video realization.
- A training-free realization that combines a persistent source-video reference with an edited proxy under an early-step conditioning schedule, with a coding agent jointly realizing reconstruction and program editing.
- Paired qualitative demonstrations spanning all eight OpenVE-Bench categories, with both the executable proxy and a matched no-proxy result visible for every main-gallery example.

2. Related Work

Video editing with generative priors. Tune-A-Video adapts an image diffusion model to a video using one-shot tuning [26]. FateZero reuses attention information obtained from the source video [21], and TokenFlow propagates diffusion features using inter-frame correspondences [7]. FLATTEN introduces optical-flow-guided attention [4], while RAVE combines frame interactions with randomized noise shuffling [13]. CoDeF represents a video through canonical content and temporal deformation, allowing image edits to be propagated across frames [20]. AnyV2V uses an edited first frame and feature injection into an existing image-to-video model [14]. These methods provide several ways to carry source information through editing. Our system adds an executable target-scene preview as a separate conditioning input to a pretrained video generator.

Explicit control for visual synthesis. Structure- and content-guided video synthesis uses distinct conditions to control these aspects of an edit [5]. ControlNet demonstrates the utility of spatial conditions such as edges, depth, and pose for image generation [29]. VideoComposer combines textual, spatial, and temporal conditions for video synthesis [24]. MotionCtrl separates camera and object motion control [25], and CameraCtrl parameterizes camera trajectories for controllable video generation [9]. VACE organizes references, video inputs, and masks within a unified creation and editing framework [12]. ReVideo combines first-frame content editing with trajectory-based motion control [18]. Our programmatic proxy can jointly render layout, camera, object behavior, and semantic changes into a video condition. The underlying generator in our implementation is the reference-conditioned MiniMax-H3 model [17].

Programs as visual representations. Holodeck generates three-dimensional environments from language and spatial constraints [28]. BlenderAlchemy uses vision-language feedback to search for graphics-editing actions [11]. The Scene Language represents scenes through programs, semantic descriptions, and identity embeddings [30]. Code-as-World investigates executable repre-

sentations for physical reasoning [23]. For video creation, Code2Video generates educational animations through executable code [2]. Code2Video-Compass evaluates programmatic video generation across text, image, and video inputs [3]; recent open-model projects also explore code-driven video creation [19]. These works establish a broad context for executable visual interfaces. Our focus is a source-grounded transition between scene programs: an observed video supplies the starting scene state, an editing instruction determines the successor state, and its render communicates the intervention to a video generator.

Code Video Model is the closest foundation for our implementation [15]. It renders executable scene programs into video previews and introduces early-step conditioning to reduce transfer of a proxy's rendered appearance. It also demonstrates scene-editing applications. We reuse these ideas and study a source-grounded setting in which observed video, rather than an existing procedural scene, is the starting point. The Video2Code–next-code–Code2Video formulation makes that reconstruction and editing interface explicit, while the original video remains a reference during synthesis.

Editing data and evaluation. InstructPix2Pix shows how generated editing pairs can support instruction-based image editing [1]. InsViE-1M constructs video-editing triplets with dedicated filtering stages and trains an instruction-following editor [27]. OpenVE-3M provides a large video-editing dataset and OpenVE-Bench, whose eight categories supply the source clips and requests used in our showcases [8]. OmniEdit-Bench and VEFX-Bench further emphasize video-specific editing coverage, instruction fidelity, and preservation assessment [6, 16]. Our present study uses selected qualitative examples. Program-conditioned data construction and broader evaluation are natural next steps for this interface.

3. Method

3.1. Video editing as a program transition

Let $V_{\mathrm{src}} = \{I_i\}_{i=0}^{N-1}$ denote the source video and e an editing instruction. We factor the editing interface into reconstruction of a source program P_0, prediction of its edited successor P_1, and video realization:

$$P_0 = \mathcal{A}_{\mathrm{rec}}(V_{\mathrm{src}}), \tag{1}$$

$$P_1 = \mathcal{A}_{\mathrm{next}}(P_0, e; V_{\mathrm{src}}), \tag{2}$$

$$V_{\mathrm{proxy}} = \mathcal{R}(P_1), \tag{3}$$

$$V_{\mathrm{edit}} = \mathcal{G}(V_{\mathrm{src}}, e, V_{\mathrm{proxy}}). \tag{4}$$

Here $\mathcal{A}_{\mathrm{rec}}$ denotes source-scene estimation and $\mathcal{A}_{\mathrm{next}}$ an instruction-conditioned program transition. The renderer $\mathcal{R}$ executes the edited program, and $\mathcal{G}$ is a frozen reference-conditioned video model. The indices of P_0 and P_1 enumerate editing states, not video frames. Both programs describe scenes that evolve over the video's own timeline.

This factorization specifies the representation and interfaces. Our current prototype supplies the source and instruction together to one coding agent, yielding $P_1 = \mathcal{A}_{\mathrm{joint}}(V_{\mathrm{src}}, e)$ directly. It jointly performs the source-grounding and editing operations, without materializing or evaluating a separate P_0 artifact. The retained program, rendered proxy, and generated output instantiate the edited branch of Fig. 2.

3.2. Video2Code: source-scene reconstruction

The purpose of Video2Code is to establish the scene state on which an edit acts. A useful source representation accounts for visible entities, their relative layout and occlusions, the camera, and motion over time. It need not recover unique three-dimensional geometry or reproduce every photographic detail. The relevant structure is the structure needed to express and inspect the requested intervention.

In our instantiation, the coding agent observes the source video and time-indexed frames while constructing a procedural Three.js scene. Object groups and geometry express the entities and their layout; camera parameters express the view; time-dependent transforms express motion and event timing. Materials, lighting, and procedural effects communicate semantic attributes and appearance. These are ordinary program constructs, rather than fields in a fixed scene-graph schema. The original composition, viewing direction, action timing, cuts, and occlusion relationships provide anchors wherever the instruction leaves them unchanged.

Source frames are available for observation only: they are not inserted into the render as image textures, video textures, background plates, or composited overlays. The procedural state therefore describes the scene explicitly. The original video supplies appearance and performance information that this approximation does not recover.

3.3. Next-code prediction: editing the scene state

Next-code prediction expresses the instruction as a transition from the source scene to an edited executable scene. Its output is a successor program, rather than a description of the desired video or a sequence of edited pixels. For an addition, the transition introduces an entity with a location, scale, and temporal behavior. For a removal, it omits the selected entity and represents the revealed context. A camera edit changes the viewing configuration; stylization and creative edits can change materials, geometry, or procedural effects. The transition may rewrite code as needed while retaining the scene relationships that remain relevant to the request.

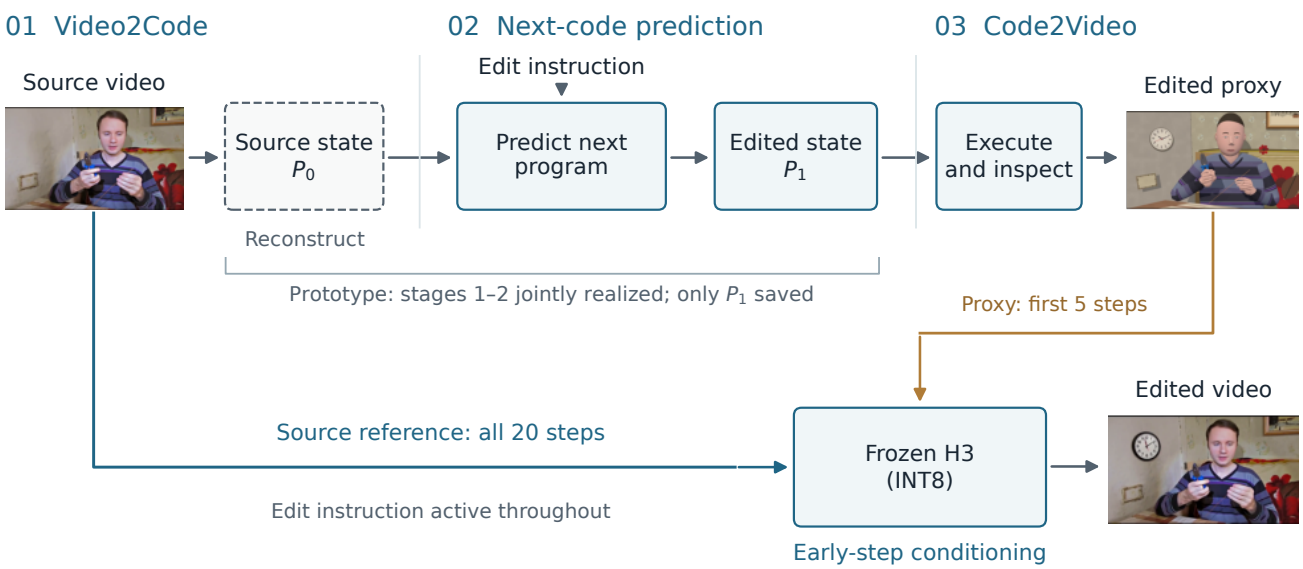


Figure 2. **Video2Code → next-code prediction → Code2Video.** At the representation level, source-scene reconstruction defines a starting program state, and the editing instruction determines its successor. Rendering the edited program provides a proxy for frozen video synthesis. In the current training-free prototype, one coding-agent session jointly realizes the first two stages and retains only the edited program. Early-step conditioning uses the proxy for the first five steps; the source reference and edit instruction remain active for all 20 steps.

This view separates the source state from the intervention conceptually, even when a coding agent constructs them jointly. For the clock in Fig. 1, the source supplies the person, wall, room layout, and action timing. The edited program adds the clock's geometry, placement, and hand animation to that context. The resulting code is an inspectable specification of the intervention.

Executable state and visual refinement. Each generated scene exposes a deterministic `seek(t)` interface that updates its camera, object transforms, visibility, and animation at time t. Evaluating timestamps out of order must produce the same corresponding states. The agent renders keyframes and contact sheets, compares them with the source observations and instruction, and revises the program. Inspection covers composition, geometry and occlusion, motion and timing, and semantic legibility. The final encoded proxy is inspected again. Execution checks verify resolution, frame count, silent output, and per-frame camera metadata. These checks establish a usable executable condition; they do not impose pixelwise equality with the source.

3.4. Code2Video: source-grounded realization

Executing P_1 at the normalized source timestamps yields the proxy $V_{\text{proxy}} = \{\mathcal{R}(P_1, t_i)\}_{i=0}^{N-1}$. The proxy makes the requested structure visible before synthesis. The source and proxy then have complementary roles: the source anchors original content and performance, and the proxy specifies the intervention. We use MiniMax-H3's reference-conditioned generation interface [17], with `<Video 1>` identifying the source and `<Video 2>` the edited structural proxy. The instruction assigns these roles and permits the requested changes.

The two conditions share a timeline. We normalize the source to 960×544 at 24 frames per second through proportional resizing and padding, and render the proxy at those timestamps. Additional frames required by the generator hold the final state of both conditions. Displayed results are trimmed back to the visible source interval.

Early-step proxy conditioning. Following Code Video Model [15], we limit proxy conditioning to the initial sampling steps. With zero-based sampling index k, total steps K, and proxy budget K_e,

$$c_k = \begin{cases} \text{Enc}(e, V_{\text{src}}, V_{\text{proxy}}), & 0 \le k < K_e, \\ \text{Enc}(e, V_{\text{src}}), & K_e \le k < K. \end{cases} \tag{5}$$

The encoder notation includes multimodal semantic conditioning and variational autoencoder (VAE) reference conditioning. We use $K = 20$ and $K_e = 5$, switching both pathways from source-plus-proxy to source-only after the

fifth step. The source VAE representation is shared between states, and the edit instruction remains active.

Sampling follows one continuous latent trajectory: switching references neither restarts sampling nor re-noises the latent. The proxy guides the developing structure early, and the remaining steps continue with the source as the visual reference. All generator and encoder weights remain frozen; program construction and visual refinement require no task-specific training.

4. Qualitative Results

4.1. Sources and implementation

We use source clips and editing requests from OpenVE-Bench [8]. The gallery presents one selected successful example from each of its eight categories, together with an introductory wall-clock example. All images come from the existing training-free early-step runs. These runs use the joint-agent instantiation of the reconstruction and next-code stages; they do not separately evaluate a pre-edit source program. The coding stage uses a multimodal coding agent, and synthesis uses a quantized MiniMax-H3 implementation with 20 Euler steps, seed 42, and proxy conditioning for the first five steps. The source remains available for all 20 steps. No additional model training or video generation was performed to construct the figures. The gallery is intended as a capability demonstration using selected outputs, rather than an aggregate benchmark comparison.

For every example, we show four time-aligned views: the source, the executable proxy, the output without a proxy, and the output with early-step proxy guidance. Both output groups use the same normalized source, main edit request, model, seed, and sampling settings. The no-proxy group uses the source throughout sampling; the guided group adds the proxy during the first five steps. The figure instructions are concise summaries of the original requests. Complete input prompts and generation settings are retained in the underlying run records; Appendix A gives reproduction details. All displayed frames preserve the full normalized composition.

4.2. Entity and spatial edits

Local addition. The seagull addition example places an animated bird in the upper-left sky. The procedural scene contains the coastline and boat as context, with the new bird placed above them. The output in Fig. 3 introduces the bird while retaining the recognizable rocky shore and sea. The clock example in Fig. 1 offers a complementary indoor addition: the proxy locates a new object relative to an existing person and wall. These examples show how an addition can be made explicit before synthesis. The no-proxy outputs introduce the requested objects but overshoot the planned scale or placement; the guided outputs follow the corresponding proxy plans more closely.

Local removal. The keyboard removal example targets the crescent-shaped keyboard surrounding a grand piano. The baseline retains this prominent keyboard ring. The proxy explicitly omits the ring while retaining the central piano, stage, and surrounding visual arrangement. The guided output follows this removal (Fig. 3), illustrating how the rendered target scene can make an omission concrete.

Local change. The person aging example replaces a woman with a silver-haired older woman while retaining her pose and position. Both output groups perform the age edit. The baseline additionally changes her white jacket into a dark dress and introduces a large necklace. The proxy-guided result retains the original outfit and phone action (Fig. 4). This comparison illustrates a more targeted intervention with fewer unrequested appearance changes.

Background change. The newsroom replacement example changes the background to a futuristic cyber command center. The baseline realizes the new environment but removes the foreground woman. The proxy includes both the speaker and the target surroundings, and the guided output retains her while changing the background. This example highlights the distinct roles of a requested background change and a foreground subject that should remain in the shot.

4.3. Camera and creative edits

Camera edit. The clock tower example requests a wide shot. The agent constructs the tower and surrounding context under the target framing. The no-proxy output remains close to the original medium view. In the guided output, the tower occupies less of the frame and more of the adjacent environment becomes visible (Fig. 5). The program makes the framing choice explicit through its camera and scene configuration. The requested wide composition applies to the edited shot; it is not an instruction to add a new reveal or a new cut.

Creative edit. The waterfront example transforms a Ferris wheel into a celestial portal. The proxy contains a luminous circular structure and its reflection, providing a spatially organized account of the transformation. The no-proxy result leaves elements of the wheel structure within the transformation. The guided result forms a coherent blue-violet portal in the waterfront setting with reflected light on the water. The example shows how procedural geometry and effects can communicate a creative target while the source supplies scene context.

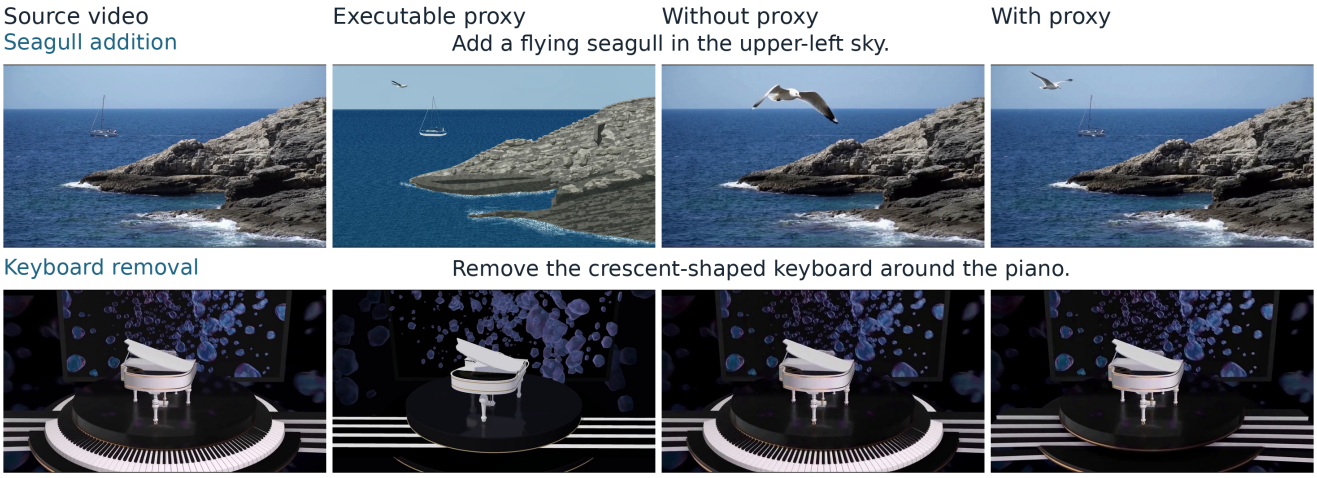


Figure 3. **Adding and removing entities.** Top: a seagull is added over the coastal scene. Proxy guidance keeps the bird closer to the planned scale and location, whereas the baseline places a much larger bird in the foreground. Bottom: the crescent-shaped keyboard around the piano is retained by the baseline and removed with proxy guidance.



Figure 4. **Targeted edits with better source-content retention.** Top: both outputs age the woman, but the no-proxy result changes her clothing; proxy guidance retains the original white jacket. Bottom: the baseline generates an empty cyber command center, while the guided result retains the foreground speaker in the new environment.

4.4. Global style and subtitle edits

Global style. The portrait example requests a Pop Art treatment of a close-up profile. The proxy renders the face with graphic contours, flat colors, and halftone details. The baseline mainly applies saturated colors to photographic texture. The guided result adopts drawn contours and halftone details while keeping the profile composition recognizable (Fig. 6). Here, appearance changes are part of the instruction. The source-retention guidance is explicitly subordinate to those requested changes.

Subtitle edit. The aerial landscape example removes subtitles from the top of the frame. The no-proxy output retains the text band. The proxy depicts the scene without text, and the guided output removes the band while retaining the recognizable fields, mountains, and cloudy sky. The example uses the video generator for the final completion; no separate text-erasure or compositing operation is applied to the displayed output.

Temporal inspection. The examples are generated as videos rather than independent edited frames. Appendix Fig. 7 provides additional timestamps for the seagull and portal examples. These sequences expose the continuing edit and its relation to the original shot.



Figure 5. **Reframing and creative transformation.** Top: the no-proxy output remains close to the source framing; the guided result realizes the requested wider shot of the clock tower. Bottom: the baseline blends a portal effect with the original wheel structure; the guided output forms a distinct celestial portal with a corresponding reflection.



Figure 6. **Stylization and subtitle removal.** Top: the portrait is rendered in a Pop Art aesthetic. The baseline mainly changes photographic colors, whereas the guided output adds graphic outlines and halftone detail. Bottom: the baseline retains the subtitle band, while proxy guidance removes it from the aerial landscape.

5. Discussion and Future Work

A controllable intermediate. The executable program exposes the scene plan used to condition the video model. Object placement, camera configuration, and temporal behavior can be checked in its render and revised in code. This visibility is useful even when the final appearance is produced by a generative model. The source video and the rendered plan provide complementary information within one interface.

From a training-free system to editing data. The current implementation records examples of the form $(V_{\text{src}}, e, P_1, V_{\text{proxy}}, V_{\text{edit}})$. Separately materializing the reconstruction stage would extend these to $(V_{\text{src}}, e, P_0, P_1, V_{\text{proxy}}, V_{\text{edit}})$, exposing the program transition itself. Such data could support video-to-program reconstruction and instruction-conditioned next-code prediction alongside video synthesis. The current runs retain P_1; collecting and validating the paired source programs remains future work. Programs can supply object, camera, and temporal information alongside the rendered condition. Combining these signals with independently screened edited videos could support training a model to distinguish the source's content from the proxy's requested intervention. Data curation and filtering would be central to this process, as demonstrated in instruction-based image

and video editing [1, 8, 27]. Generated targets would be quality-filtered pseudo-labels. Any resulting improvement would need to be established through training and held-out evaluation.

Scope and next steps. The current program is an approximate, task-conditioned account of a scene. It does not recover uniquely determined geometry or all of the source's appearance and performance. The generator can also alter details outside the intended edit. Our selected showcases establish feasibility for the displayed requests; they do not measure success frequency across the benchmark. Further work includes broader quantitative paired evaluation, human preference studies, and measurements of the cost of program construction. Editing-specific benchmarks provide useful protocols for separating instruction accuracy, preservation, realism, and consistency [6, 16]. Evaluating reconstruction, program transitions, and final video quality separately would help identify the source of editing failures. Learning these interfaces and learning to use the source and proxy jointly are promising extensions of the present system.

6. Conclusion

We formulated video editing as Video2Code, instruction-conditioned next-code prediction, and Code2Video. The source scene establishes an executable state, the editing instruction defines its successor, and a rendered proxy makes the intervention visible. Our training-free prototype jointly realizes reconstruction and program editing with a video-aware coding agent. A frozen video model combines that proxy with a persistent source reference, using proxy conditioning only during the early sampling steps. Selected examples across the eight OpenVE-Bench categories illustrate the range of edits expressible through this interface.

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A. Implementation and Figure Provenance

Program construction. Our implementation adapts the Code Video Model program-and-render workflow to observed source videos. The three-stage formulation separates source reconstruction from prediction of the edited program at the representation level. The existing prototype gives the source and instruction jointly to the agent and serializes only P_1, together with its rendered proxy. It does not run two independently checkpointed coding modules or save a standalone pre-edit P_0 reconstruction. The coding stage uses the Codex CLI with GPT-6 Astra and the `xhigh` reasoning setting. Each case has its own procedural Three.js workspace, source-frame index, rendering helper, and task contract. The agent renders and directly inspects keyframes and encoded-video contact sheets. The renderer evaluates frame i at time $i/24$ through a deterministic `window.reconstruction.seek(t)` interface. A camera-state accessor exports position, quaternion, and field of view after the corresponding time has been evaluated. This metadata is used to validate the procedural output. It is not supplied as a separate camera-control tensor to the video model.

Video synthesis. We use the INT8 H3 implementation from the existing early-step experiment, with a quantized Qwen visual-language encoder. Video-VAE and audio-VAE computation use the established FP16 and FP32 paths, respectively. Sampling uses 20 Euler steps, seed 42, video/audio shifts of 12/3, and the unchanged sigma schedule of that experiment. Both multimodal semantic conditioning and VAE reference conditioning switch from source-plus-proxy to source-only after the fifth step. There is one continuing audiovisual latent trajectory. The source reference latent is shared across the full and reduced conditioning states. All parameters remain frozen.

For throughput, the existing experiment used eight resident single-GPU workers to generate independent cases. This is case-level parallelism; one video does not require the eight workers to participate in a single distributed forward pass. Figure construction only decodes existing videos and does not run H3 again.

Timeline and audio. Source videos are normalized to 24 FPS and proportionally resized and padded to 960×544. If the normalized source has N frames, alignment uses the smallest admissible $N' \geq N$ satisfying $N' = 17m+5$. The additional $N' - N$ frames hold the final source and proxy states. The displayed edited videos are trimmed to the original N-frame interval. Proxies are silent. Generated audio is retained in the original experiment outputs; the present figures and qualitative descriptions concern visual editing.

Showcase selection. The figures use outputs from the `openve_h3_int8_early_step_20261031` experiment. Existing automated judgments were used to shortlist candidates, followed by direct visual inspection of the source, proxy, and both output groups. Selection targets successful illustrations of the requested edits. It is not a random benchmark sample. The main gallery illustrates seagull addition, crescent keyboard removal, person aging, newsroom replacement, clock tower reframing, a celestial portal, Pop Art stylization, and subtitle removal. The introductory figure shows wall clock addition. The same generated output is used wherever a case appears more than once.

Image construction. The supplied figure script decodes exact frame indices from the published source, proxy, and edited MP4s. It preserves the full normalized frame and aspect ratio, then arranges the images with external labels. No image-generation model, retouching, or replacement of visual content is used in making the figures. The accompanying `research/showcase_provenance.json` records video hashes, sampled frame indices, timestamps, prompt hashes, and figure hashes. Figure edit summaries are paraphrases for readability; they are not additional instructions supplied during generation.

B. Conditioning Instructions

Coding-agent task contract. The task provides the source video, timestamped observations, the benchmark edit request, and exact output metadata. Its core requirements are to build a deterministic, fully procedural, editable Three.js scene; implement the requested edit; and retain unedited composition, geometry, occlusion, motion, cuts, and timing. Source imagery is available for observation only. External visual assets and source imagery are not used as rendered textures, plates, overlays, or masks. The contract also requires visual refinement passes covering composition, geometry and occlusion, motion and timing, and semantic legibility. The final encoded video is inspected before it is accepted as a condition.

Reference roles in H3. The no-proxy comparison uses the same source and main editing request, with a reference-role block that identifies only the source video. The proxy-guided group adds the second reference as described below. The original edit request is followed by the source-reference guidance below. This is the reference-role block used in the existing proxy-conditioned showcase runs; it introduces no shot-by-shot target-video description. The numbered video references correspond to their actual input order.

> `source_reference_guidance: <Video 1>` is the source video to edit. Preserve its object identities, appearance, color palette, lighting, textures, mate-

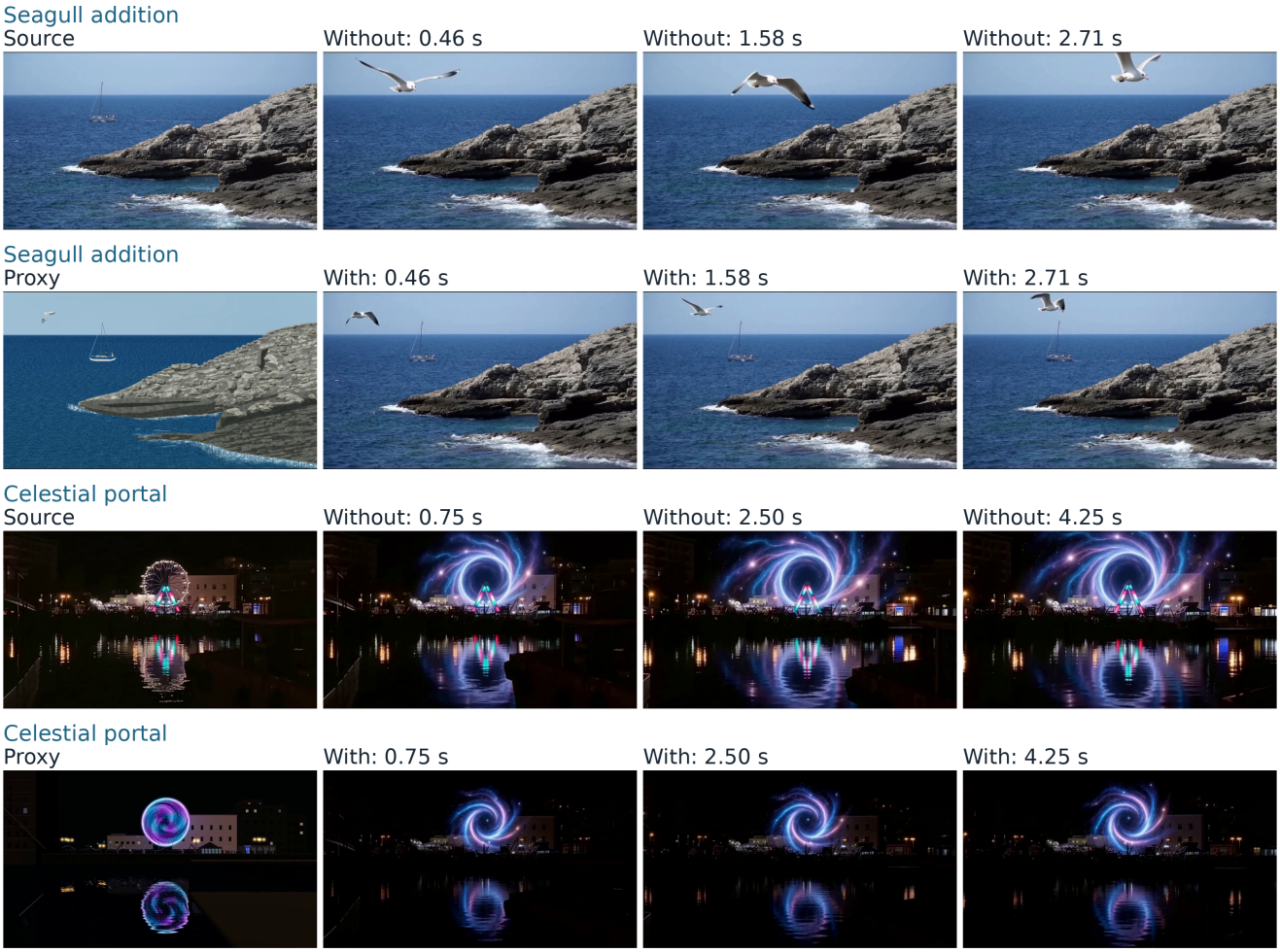


Figure 7. **Additional temporal views of existing showcase videos.** Each case has two rows: a source frame followed by three no-proxy timestamps, and a proxy frame followed by the matching guided timestamps. Top pair: seagull addition. Bottom pair: the celestial portal. All outputs are the same videos used in the gallery.

> rial treatment, rendering medium, and overall aesthetic, as well as its composition, poses, spatial layout, motion, camera, shot transitions, and timing, except where the main edit prompt requires a change. `<Video 2>` is the structural proxy video showing the requested edit. Use `<Video 2>` as structural guidance for the requested edit; do not transfer its rendered appearance to the output. Follow the main edit prompt for the requested changes.

The edit request and reference-role text stay in the conditioning prompt throughout the trajectory. After step five, the proxy's visual reference and its associated multimodal conditioning are withdrawn as specified in Eq. (5). For requested stylization or appearance edits, the explicit exception in the preservation instruction allows the intended change. Source preservation is a conditioning objective rather than a pixelwise compositing constraint.